

Multi-Agent Deep Reinforcement Learning for Vehicular Resource Allocation: A Comparison Study of Different Agent Cooperation Levels

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Abstract—The spectrum sharing design (such as the underlay sidelink configuration in cellular networks) for vehicle-to-vehicle (V2V) and vehicle-to-network (V2N) links can improve the efficiency of spectrum utilization for vehicular networks. However, differentiated service requirements for V2V and V2N links and the greater complexity of interference management make it challenging for conventional centralized learning-based methods in the face of real-world deployment. This motivates the exploration of multi-agent deep reinforcement learning (DRL) solutions, while existing research has predominantly focused on cooperative setting that causes large communication overheads for real-world deployment. This paper proposes a decentralized resource allocation scheme for joint spectrum sub-band and V2V transmission power selection that utilizes federated learning (FL) to formulate a mixed setup to avoid frequent communication and enable easy model sharing, termed FL-MX for the design of FL and mixed DRL agent cooperation level. Different levels of cooperation among agents are also explored by implementing different state-of-the-art (SOTA) DRL-based resource allocation algorithms on the developed standard-compliant simulation platform. Through extensive simulations, it is observed that the proposed FL-MX manages to converge effectively and can match the best SOTA performance, gaining a 22.55% and 9.5% improvement in the V2V payload delivery ratio and V2N throughput compared to the random baseline. Such results make it promising for future research to investigate multi-agent DRL designs at a mixed agent cooperation level for high-performing network decision solutions with less frequent communication demands.

Index Terms—Federated learning (FL), multi-agent deep reinforcement learning (DRL), resource allocation, vehicular network.

I. INTRODUCTION

THE growing demand for intelligent transportation systems (ITS) has highlighted the importance of vehicular networks, which facilitate communication to ensure road safety, improve traffic efficiency, and provide on-board infotainment services [1]. Vehicle-to-vehicle (V2V) and vehicle-to-network (V2N) are two essential communication types in vehicular networks, which are utilized to support various applications [2]. Real-time safety-critical messaging between nearby vehicles is

accomplished via V2V links, with strict latency and reliability requirements. On-board infotainment services through the Internet are typically supported by V2N links that require high capacity and large bandwidth between vehicular user equipments (VUEs) and base stations (BSs) or roadside units (RSUs).

With a scarce frequency spectrum and a significantly increasing number of VUEs, wireless resources must be efficiently managed to support various V2X communication services. This motivates standardization, such as the underlay sidelink architecture [3], [4] in Cellular V2X (C-V2X) [5], a promising technology that is currently under active development. In C-V2X, the underlay architecture allows V2V and V2N links to co-exist on a shared resource pool to enhance spectrum efficiency.

However, the diverse service requirements between V2V and V2N, along with the complex interference management requirements introduced by the underlay architecture, pose a significant challenge for conventional resource allocation approaches to satisfy, necessitating efficient resource allocation for proper spectrum sub-band allocation and transmission power control [6], [7]. This motivates the use of machine learning in recent years [8] for resource allocation optimization, especially the proposal of single-agent and multi-agent deep reinforcement learning (DRL) algorithms that utilize artificial neural networks (ANNs) to various learning tasks for improved task-specific performance.

Nonetheless, centralized single-agent DRL and traditional multi-agent DRL solutions face significant limitations in V2X networks [9]. Centralized single-agent setups rely on a network controller for all decisions, which can create bottlenecks in large, dynamic environments due to high data aggregation demands and continuous updates, limiting scalability and increasing latency. The existing multi-agent DRL requires frequent direct information sharing among agents, leading to the same issue as the single-agent setup, while making it challenging for new agents to join without retraining the learning models. Federated learning (FL) overcomes these challenges by aggregating individual model updates into a global model, reducing the need for continuous data exchange and preserving agent privacy [10]. Newly joined agents can directly use the global model, reducing deployment costs. In V2X, FL's adaptable communication intervals and local decision-making also lower communication overhead and latency, while ensuring scalability and data security, making it ideal for decentralized spectrum and power allocation [9], [11].

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Until now, existing research has mostly focused on the design of cooperative agents and centralized training setups for multi-agent DRL-based solutions, regardless of FL utilization. This introduces large communication overheads, as information must be frequently sent to the central controller or shared between agents to train and execute decision-making algorithms. Compared to a cooperative setting, a mixed setting with self-interested agents does not require direct information sharing between agents to attain a common reward, and thus can reduce frequent communication during training. However, a fully distributed information structure in multi-agent DRL at a mixed agent cooperation level may face the extreme case of independent learning, which can cause non-convergence in the learning process [12].

It should be noted that although C-V2X serves as the backbone technology in this work, the above research gap is about the broader challenge of decentralized resource allocation in vehicular networks that can also be adapted to other communication protocols that support spectrum-sharing between V2V and V2N links, such as the CAR 2 CAR Communication Consortium specified in IEEE 802.11bd [13].

A. Objectives and Contributions

Non-cooperative multi-agent DRL solutions for resource allocation have received less attention than the cooperative setup due to their inherent challenges of potential instability in learning convergence, and thus require further investigation. FL is a promising technique that enables knowledge-sharing among agents in multi-agent DRL without direct information transmission through global model aggregation. Therefore, FL can preserve user data privacy while providing a convenient model-sharing mechanism for enhanced scalability. As ANN models are transmitted less frequently, FL becomes a promising framework for developing multi-agent DRL-based decentralized resource allocation schemes with non-cooperative agents to avoid frequent communication.

Consequently, a decentralized joint sub-band and transmission power allocation scheme for cellular vehicular networks is proposed for spectrum-sharing V2V and V2N links for C-V2X with an underlay sidelink configuration. The proposed scheme aims to maximize the successful delivery ratio of the V2V safety payload while also maximizing the overall V2N rate in the system. The overarching goal of this work is to explore the performance of schemes with different agent cooperation levels and the deployment costs of the different schemes; hence, to pave the way for high-performing, practical, and low-cost multi-agent DRL-based resource allocation schemes for C-V2X development. The main contributions of this work are:

- A Federated DRL-based solution proposed for joint power & sub-band resource allocation for sub-6 GHz underlay cellular vehicular network with mixed agent cooperation setups.
- The latest standard-compliant (3GPP TR 37.885) simulation testbed developed with Python for joint power & sub-band allocation experiments.

- Performance evaluation of different agent cooperation levels through comparing different state-of-the-art DRL-based solutions. The empirical results demonstrate the effectiveness of a mixed cooperation setup for multi-agent DRL in resource allocation.

The remainder of this paper is organized as follows: A review of related work is covered in Section II; Problem formulation in Section III; The proposed DRL-based algorithm and the federated DRL resource allocation scheme in Section IV, followed by the simulation setup and performance evaluation results in Sections V and VI, respectively, culminating in the conclusion and future research insights in Section VII.

II. RELATED WORK

Evolved from the same system setup as in traditional solutions, the centralized learning setup pioneered the adoption of DRL in resource allocation decision making in vehicular networks. In the centralized learning setup, the network controller takes measurements from VUEs and makes resource allocation decisions, such as the work of Ye et al. [14]. However, a centralized learning setup in vehicular networks has obvious limitations, as the central agent is under heavy computation demands with large communication overheads for both training and execution. Therefore, this design paradigm usually suffers from scalability issues in large networks. These drawbacks motivated the multi-agent DRL setup to directly exploit local information at each vehicle to distribute computation tasks and explore the environment better.

A. Multi-Agent DRL in V2X Resource Allocation

The pioneer in this research direction for exploring multi-agent DRL-based V2X resource allocation may be Liang et al. [15]. In their work, a multi-agent deep Q-learning for joint sub-band and V2V transmission power selection in a spectrum-sharing cellular V2X was developed. Each V2V link was modeled as an agent, cooperating with other vehicular agents to maximize the V2V payload delivery for all V2V links and the V2N sum transmission rate in the local communication network. This idea was extended to continuous power allocation in the latest work of Cai et al. [16], in which a DRL algorithm based on proximal policy optimization with the reward function design was proposed, also considering the number of successfully delivered V2V packets. Similarly, Gu et al. [17] proposed a resource allocation based on a multi-actor attention-critic algorithm in a 3GPP standard-compliant sensing-based scheduling setup. All VUEs shared their state transition observations with the RSU for centralized policy training [17].

To further enhance the generalization capability of DRL models, hierarchical DRL solutions for joint sub-band and V2V transmission power selection were also developed, such as the work of Yuan et al. [18] and Sohaib et al. [19]. In these works, meta-learning algorithms were implemented to adaptively tune their proposed resource allocation schemes in response to different communication scenarios. However, two

DRL algorithms (deep Q-learning and deep deterministic policy gradient (DDPG) [20]) were utilized in [18] for spectrum allocation and continuous power control, respectively, while a TD3-based algorithm [21] was proposed in [19] as the resource allocation optimization scheme.

Although satisfactory performance can be achieved, the above solutions have difficulties with online model updates after deployment, as only the agents from the original training phase can have a working DRL model. Sharing such a trained model with a newly joined agent is not a trivial task.

B. Federated DRL in V2X Resource Allocation

Federated learning is a decentralized learning architecture that uses data distributed among different local learners. As introduced above, FL provides an effective mechanism for the ANN model and knowledge sharing while protecting the privacy of participants' information. For resource allocation, Zhang et al. [22] proposed a federated multi-agent DRL solution for each VUE cluster in a local area for sub-band and V2V power allocation under a cooperative setup via a shared reward design. A graph-based VUE clustering was further proposed for efficient model sharing. In comparison, Li et al. [23] proposed a dueling double deep Q-network for channel and power allocation. Their design of the reward function considered an additional penalty element for high interference and delays. Rasheed [24] also used the deep Q network for the joint resource allocation task while exploring an asynchronous federated aggregation method. A shared reward function was designed that considered channel capacity, reliability, and latency requirements. This still holds for the recent work of Xu et al. [25], which also utilized an FL-enhanced multi-agent DRL using a policy gradient-based algorithm with a shared reward among vehicular agents during the DRL training phase. To the best of the authors' knowledge, only recent research by Parvini et al. [26] has begun to investigate the allocation of V2X radio resources using decentralized federated reinforcement learning, focusing on high-speed queuing scenarios.

C. Research Gap and Motivation of This Work

Despite much attention paid to multi-agent DRL-based decentralized resource allocation (with or without FL) in vehicular networks, existing solutions focused on the setting of cooperative learning and were typically configured in a "centralized training, distributed execution" manner. This approach has been preferred because all agents aim to achieve a common objective of optimization without being self-interested and competing for radio resources. However, this configuration requires frequent information sharing among agents during the training stage to formulate a global state observation or shared instantaneous reward for all agents. This causes large communication overheads, especially combined with the requirement of sending ANN models with the introduction of FL model aggregation.

Consequently, it leads to the motivation of this work to explore different levels of agent cooperation to fully utilize the FL architecture. In addition, as the reviewed research still carried

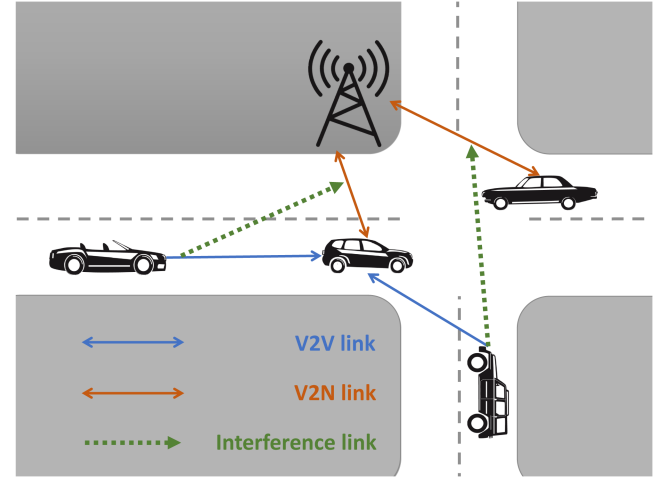


Fig. 1. A vehicular network with spectrum sharing V2N & V2V links.

out the experiments under the previous C-V2X standards, simulations implemented to be compliant with the latest standards that resemble a 5G NR-V2X network also become necessary for algorithmic performance evaluation.

III. SYSTEM MODEL AND PROBLEM FORMATION

The considered cellular vehicular network system consists of one BS and multiple VUEs, as shown in Fig. 1, forming J V2N and K V2V links in the network. As discussed in 3GPP's Release 15 and 16 specifications for cellular V2X enhancement [27], V2N and V2V links provide simultaneous support for high data rate infotainment services and reliable safety message sharing for essential safety applications and advanced driving services, respectively. Note that mode selection for safety messaging is disabled; that is, VUEs transmit safety messages only through V2V links, and V2N links are only for information services. Specifically, this work considers Mode 2 for the C-V2X architecture as a distributed mechanism for the spectrum selection of V2V links. Each vehicle can autonomously select radio resources for its V2V link instead of relying on the BS to allocate resources centrally. Define the set of V2N and V2V links as $\mathcal{J} = \{1, 2, \dots, J\}$ and $\mathcal{K} = \{1, 2, \dots, K\}$, respectively, while the total bandwidth is divided into M orthogonal spectrum sub-bands, forming the set $\mathcal{M} = \{1, 2, \dots, M\}$. Next, denote $\rho_{j,m}$ and $\rho_{k,m}$ ($\rho_{j,m}, \rho_{k,m} \in \{0, 1\}$) for the sub-band selection decisions to V2N and V2V links, that is, $\rho_{j,m} = 1$ or $\rho_{k,m} = 1$ if the j -th V2N link and k -th V2V link select the m -th sub-band. Each V2V link can select one sub-band at a time, while multiple V2V and V2N links can share the same sub-band to improve spectrum utilization. It is further assumed that the orthogonal spectrum sub-bands are uniformly preassigned to each V2N link with fixed transmission power, i.e. the j -th V2N link always occupies sub-band $\bar{m}_j$ or $\rho_{j,\bar{m}_j} = 1$, $\bar{m}_j = \text{mod}(j, M)$.

The channel power gain consisting of both large- and small-scale fading components is considered. The channel gains for the j -th V2N link over sub-band $\bar{m}_j$ and the k -th V2V link on sub-band m are represented by $h_{j,\bar{m}_j}$ and $g_{k,m}$, respectively. The interference to the j -th V2N link from the k -th V2V link

over the $\bar{m}_j$ -th sub-band is gk . The signal-to-interference plus noise (SINR) ratio received from the j -th V2N link, and that of the k -th V2V link over sub-band m can be expressed as:

$$SINR_{j,\bar{m}_j}^{v2n} = \frac{p^{v2n} h_{j,\bar{m}_j}}{I_{j,\bar{m}_j} + \sigma^2} \quad (1)$$

$$SINR_{k,m}^{v2v} = \frac{p_{k,m}^{v2v} g_{k,m}}{I_{k,m} + \sigma^2} \quad (2)$$

where p^{v2n} and σ^2 stand for the fixed V2N transmission power and the noise power, respectively, while $p_{k,m}^{v2v}$ represents the transmission power of the k -th V2V link on sub-band m . $I_{j,\bar{m}_j}$ and $I_{k,m}$ denote the total interference power to V2N link j over sub-band $\bar{m}_j$ and V2V link k over sub-band m , calculated as:

$$I_{j,\bar{m}_j} = \sum_{j'=1}^J p^{v2n} h_{j',\bar{m}_j} + \sum_{k=1}^K \rho_{k,\bar{m}_j} p_k^{v2v} g_{k,\bar{m}_j}; \quad j' \neq j, \bar{m}_{j'} = \bar{m}_j \quad (3)$$

$$I_{k,m} = \sum_{j=1}^J p^{v2n} h_{j,\bar{m}_j} + \sum_{k'=1}^K \rho_{k',m} p_{k'}^{v2v} g_{k',m}; \quad k' \neq k, \bar{m}_j = m \quad (4)$$

where $h_{j',\bar{m}_j}$ and $g_{k,\bar{m}_j}$ represent the interfering channel gains at the j -th V2N receiver from the transmitter of the j' -th V2N link and the k -th V2V link over sub-band $\bar{m}_j$. Similarly, $h_{j,\bar{m}_j}$ and $g_{k',m}$ stand for the interfering channel gains at the k -th V2V receiver from the j -th V2N link and k' -th V2V link over the m -th sub-band.

With the above definitions, the capacity of the j -th V2N link and k -th V2V link can then be obtained by (5) and (6), according to Shannon capacity theory:

$$C_j^{v2n} = W \log(1 + SINR_{j,\bar{m}_j}^{v2n}) \quad (5)$$

$$C_k^{v2v} = \sum_{m=1}^M \rho_{k,m} \cdot W \log(1 + SINR_{k,m}^{v2v}) \quad (6)$$

where W is a constant that represents the bandwidth of the sub-bands. Note that since V2V links can only choose one sub-band at a time step, (6) essentially finds sub-band m such that $\rho_{k,m} = 1$ since $\rho_{k,m'} = 0$ where $m' \neq m$.

As discussed above, the V2N and V2V links need to satisfy differentiated service requirements. The V2N links are designed to provide information services that require a high data rate, forming the clear goal of maximizing the V2N sum rate $\sum_{j=1}^J C_j^{v2n}$.

In comparison, V2V links focus on the reliable transmission of safety-critical messages, which leads to the network goal of maximizing the average successful transmission probability among all V2V links. For the k -th V2V link, the successful transmission of payload u is represented by a binary value

$\zeta_{k,u} \in \{0, 1\}$, calculated by the following Boolean expression:

$$\zeta_{k,u} = \sum_{t=t_u}^{t_u+t_0/\Delta t} \Delta t \cdot C_k^{v2v}(t) \geq B_u \quad (7)$$

where B_u is the size of the u -th payload for V2V link k . t_u and Δt are the starting time of payload u ' transmission and the duration of a time step, respectively. t is the time step index corresponding to the transmission time and t_0 is the maximum delay tolerance for V2V payload transmission. Based on (7), the delivery of V2V payload u for V2V link k is successful ($\zeta_{k,u} = 1$) only if the delivery time consumption does not exceed t_0 , and unsuccessful ($\zeta_{k,u} = 0$) otherwise. Note that t_u of each V2V link to transmit payload u is the same for the synchronous transmission setup considered in this work, and may differ if an asynchronous setup is considered. Therefore, the successful transmission probability of V2V link k during a given transmission period can be expressed as follows, assuming that $N_{payload}$ V2V payloads have been transmitted during this period:

$$\eta_k = \frac{\sum_{u=1}^{N_{payload}} \zeta_{k,u}}{N_{payload}} \quad (8)$$

Following the above definitions, the optimization problem for the considered spectrum-sharing scenario can be formulated as follows at each resource allocation time step:

$$\max_{\hat{\rho}, \hat{P}^{v2v}} E = \frac{\sum_{k=1}^K \eta_k}{K} \quad (9a)$$

$$\text{s.t. } \max \Lambda = \sum_{j=1}^J C_j^{v2n}, \quad (9b)$$

$$\sum_{m=1}^M \rho_{k,m} \leq 1, \forall k \quad (9c)$$

$$0 \leq p_{k,m}^{v2v} \leq P_{\max}^{v2v}, \forall k \quad (9d)$$

where $\hat{\rho} = \{\rho_{1,1}, \dots, \rho_{k,m}, \dots, \rho_{K,M}\}$ is the set of sub-band selection indicators, while $\hat{P}^{v2v} = \{p_{1,1}^{v2v}, \dots, p_{k,m}^{v2v}, p_{K,M}^{v2v}\}$ is the set of V2V power allocation indicators, and $P_{\max}^{v2v}$ is the maximum transmission power of a V2V transmitter. For power control, this work considers discrete power control following the reasoning of [28] and assumes that the transmit power of VUEs has N_p levels.

The resource allocation problem formulated in (9) is a multi-objective problem that aims to maximize both the V2V payload delivery probabilities and the V2N sum rates at the same time. Furthermore, since the problem involves sequential decision making in multiple time steps for resource allocation, this problem is NP-hard [18] and is highly challenging to solve using a conventional optimization solution, especially considering the time limit set to produce resource allocation decisions for real-time applications.

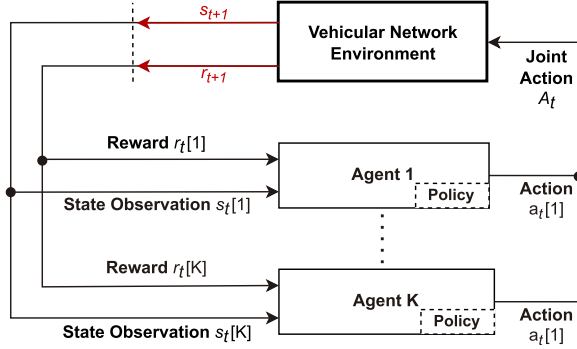


Fig. 2. The Markov Game model for multi-agent reinforcement learning [29].

IV. PROPOSED DRL-BASED RESOURCE ALLOCATION ALGORITHM

Delving into the essence of joint resource allocation, the optimization problem considered can be modeled as a Markov Game (also known as a Stochastic Game) [29]. The Markov Game framework generalizes the Markov Decision Process [30] to a multi-agent configuration. As described in Fig. 2, multiple agents interact with the environment and other agents simultaneously in a Markov Game, where the state transitions and the cumulative return of an agent depend on the actions of all participating agents. This framework is well-suited to the considered decentralized resource allocation problem, where vehicular agents decide their sub-band and transmit power selections at every time step. Based on this framework, DRL is selected as the optimization methodology because it efficiently handles complex decision-making problems in large state and action spaces, which is inherent in resource allocation for vehicular networks. Consequently, an FL-enhanced decentralized DRL solution is proposed for the original optimization problem in (9).

The remainder of this section covers the design of the three key elements of the Markov Game model, i.e., state space, action space, and immediate reward, while also proposing the DRL algorithm for resource allocation. Without losing generalization, it is assumed that the scheduling period of V2V pairs is set to be 1 time step for the whole set of V2V pairs.

A. State and Observation Space

For the considered problem, the true environmental state at any time step $\mathcal{Z}_t$ may include global channel information, all agent behaviors, and other related information. However, individual agents cannot collect such global details and can only receive state information by observation, forming an observation space denoted $\mathcal{S}_t$ for all agents.

To reduce communication overhead, the state observation of an agent at each time step t should only include local information at each vehicular receiver, which consists of the SINR of the V2V link detected in each sub-band $\overline{\text{SINR}}_t$, the selection of sub-band and transmission power for the last time step $\bar{\rho}_{t-1}$ and $\bar{p}_{t-1}^{v2v}$, the remaining V2V payload B'_t , and the remaining time budget $T_{b,t}$. For N_{sb} sub-bands, $\overline{\text{SINR}}_t = \{\text{SINR}_1, \dots, \text{SINR}_{N_{sb}}\}$.

The information of $\bar{\rho}_{t-1}$ and $\bar{p}_{t-1}^{v2v}$ is designed to be represented by one-hot encoding [31] to avoid potential confusion with a decimal SINR value. For example, if an agent selects sub-band 2 with the 5 sub-bands in total, then this action after one-hot encoding becomes the vector $\bar{\rho} = \{0, 1, 0, 0, 0\}$, while $\bar{p}^{v2v}$ is also represented in this manner. B' and T_b are two scalar values. Therefore, s_t of an agent can then be formally defined as:

$$s_t = \{\overline{\text{SINR}}; \bar{\rho}_{t-1}; \bar{p}_{t-1}^{v2v}; B'_t, T_{b,t}, t\} \quad (10)$$

which has the size of $2N_{sb} + N_p + 2$, which is a constant for a defined system setup, while the observation space $\mathcal{S}_t$ is the set of s_t for all agents at the time step t .

B. Action Space

The problem formation (9) defines the actions of each agent to be the selection of the spectrum sub-band and the control of the V2V transmission power at each time step. According to the notation in Section III, such an action is defined as $\{m, p^{v2v}\}$. The spectrum naturally breaks into M disjoint sub-bands, while the V2V transmission power is defined to have N_p levels as previously defined, such that $m \in \mathcal{M}$ and $p^{v2v} \in \{0, \dots, \frac{P_{\max}^{v2v}}{N_p - 1}\}$, forming an action space sized $M \cdot N_p$ representing all possible $\{m, p^{v2v}\}$ combinations. Therefore, the action space is defined as $a_t = \{1, \dots, MN_p\}$, with each action in a_t associated with a combination of spectrum sub-band and power selection.

C. Reward Design

Reinforcement learning is advantageous for decision optimization by designing a reward function to represent difficult-to-optimize objectives and constraints with a high degree of flexibility. After an agent takes an action based on its policy, it receives an immediate reward generated by the environment that indicates the performance of this decision execution.

Following the optimization objective in (9), an optimal result for each V2V agent maximizes the probability of successful delivery of a V2V payload within the time tolerance while sacrificing the V2N transmission rate as little as possible, which is the core concept for the reward design.

Because every decision must be based solely on the local observations of an agent, the immediate reward is defined as follows:

$$r_t = \begin{cases} C_t^{v2v} - \Delta B, & \text{if } B'_t > 0 \\ C_t^{v2v} + C_t^{v2n}, & \text{else} \end{cases} \quad (11)$$

Here, ΔB is the average transmission rate required to deliver a V2V payload of size B under the specified network configuration, while C_t^{v2v} and C_t^{v2n} are the transmission rate of the agent's corresponding V2V link and V2N link at the same vehicle receiver, consistent with the notions in Section III. The first branch of (11) normalizes the V2V rate by ΔB , giving the agent an explicit baseline to meet the V2V payload-latency constraint. Once the payload has been delivered ($B'_t \leq 0$), the reward switches to the second branch, which maximizes the V2N sum rate while retaining a V2V component. Because the second branch yields a substantially higher reward, it further

incentivizes the agent to complete the V2V payload transmission promptly.

D. DRL Algorithmic Implementation

To solve the Markov Game formulated for the considered resource allocation optimization problem, this work utilizes the well-known deep Q-network (DQN) method originally proposed by Mnih et al. [32]. To be specific, this work develops the algorithm based on the double deep Q-network (DDQN) variant of the DQN [33]. This algorithm will also be the backbone for all considered benchmarks as introduced in Section V, providing a standardized algorithmic flow for a more well-controlled performance comparison between different levels of agent co-operation.

For DDQN, the calculation of the Q value is shown in (12) according to the Bellman equation [30]:

$$Q(s_t, a_t; \theta_q) = r_t + \gamma Q(s'_t, \arg\max_{a'} Q(s'_t, a'; \theta_q); \theta'_q) \quad (12)$$

where s_t and a_t represent the current state and the current action taken at the time step t , respectively, while s'_t and a' represent the resultant state and the action taken in that state. In addition, r_t and $\gamma \in [0, 1]$ represent the received reward and the discount factor, while θ_q and θ'_q are the training and target ANN, respectively. DDQN introduces θ'_q to calculate the predicted value $Q(s'_t, a')$, which in the original DQN is calculated by the same θ_q . During the DDQN training process, only θ_q is updated following each training iteration, while θ'_q only periodically synchronizes with θ_q by copying all parameters from θ_q to keep the information up to date. This design greatly reduces the potential instability during training compared to the original DQN and thus increases the convergence probability. In addition, the technique named Experience Replay [34] is also introduced to speed up the learning process by utilizing data collected at different stages.

The implementations of DDQN for the proposed solution are summarized in Algorithm 1. The ϵ -greedy action selection method is used during training so that an agent selects a random action with probability ϵ instead of always greedily following its learned policy to avoid the local optimum and ensure a comprehensive search for the optimal policy. This is also the configuration for all benchmarks as introduced in Section V-B. Note that the presented algorithm is for one time step t , which, by calling upon multiple iterations, can represent the complete training phase.

E. Federated DRL-Based Resource Allocation

To enable knowledge sharing without direct information transmission, this work explores the FL setup among agents to indirectly share their local knowledge to improve the probability of convergence for agents with mixed levels of cooperation. The BS is implemented as the global model aggregator in the proposed federated DRL resource allocation scheme.

The core process of federated DRL is global aggregation that defines how local models are combined to form a global model. This work adopts the federated averaging method [35] for global aggregation. With N_{agent} V2V pairs participating in

Algorithm 1: The Proposed DDQN Algorithm.

```

/*  $t$  - current time step;  $s_t$  - state
   observation;  $T$  - last time step */

1 Function TrainingDQN( $t, s_t, T$ ):
   Data:  $\theta_q$  - Training network;
    $\theta'_q$  - Target network;  $\mathbb{B}$  - Reply buffer;
    $N_{rp}$  - Replay buffer capacity;
    $N_{mb}$  - Mini-batch size;  $\gamma$  - Discount factor;
    $N_f$  - Update frequency of  $\theta'_q$ ;
    $\epsilon$  - Random action probability.

2    $a_t \leftarrow \begin{cases} \text{a random action,} & \text{with } \epsilon \\ \arg\max_a Q(s_t, a; \theta_q), & \text{with } 1 - \epsilon \end{cases}$ 
3   Execute  $a_t$  and observe  $s'_t$  and  $r_t$ ;
4   Store ( $s_t, a_t, s'_t, r_t$ ) in  $\mathbb{B}$ ;
5   if  $N_{rp}$  is reached for  $\mathbb{B}$  then
6     Delete the oldest sample in  $\mathbb{B}$ ;
7   end
8   if  $\mathbb{B}$  reaches  $N_{mb}$  samples and  $t = T$  then
9     Sample a mini-batch of ( $s, a, s', r$ ) from  $\mathbb{B}$ ;
10    foreach ( $s, a, s', r$ ) sample do
11      // Construct target value
12      if  $s'$  is the terminal state then
13         $y \leftarrow r$ ;
14      end
15      else
16         $\hat{a} \leftarrow \arg\max_{a'} Q(s', a'; \theta_q)$ ;
17         $y \leftarrow r + \gamma Q(s', \hat{a}; \theta'_q)$ ;
18      end
19      Calculate loss:  $\|y - Q(s, a; \theta_q)\|^2$ ;
20    end
21    Update parameters of  $\theta_q$ ;
22  end
23  if  $N_f$  is reached then
24    /* Update target network */
25     $\theta'_q \leftarrow \tau \theta_q + (1 - \tau) \theta'_q$ ;
26  end

```

the federated DRL, the ANN weights of the global model θ_g can be updated by:

$$\theta_g \leftarrow \frac{1}{N_{agent}} \cdot \sum_{id=0}^{N_{agent}} \theta_{agent_{id}} \quad (13)$$

where $\theta_{agent_{id}}$ represents the ANN weights of the local model of $agent_{id}$. Note that this (13) is a simple global aggregation method that assumes the same aggregation weight for every participant utilized in this work. As the focus of this work is to explore how a federated DRL with mixed agent cooperation level performs compared to a cooperative DRL setup, we assume that the training scenario contains a predefined set of participants without newly added and dropped participants. Moreover, note that ensuring the models' successful transmission and model compression for communication overhead reduction is beyond the scope of this work. Therefore, the adopted aggregation method is suitable for the scenario considered.

Algorithm 2: Proposed FL-MX Allocation Scheme.

Initialize: BS aggregator distributes the global model to each participating agent
Data: θ^g - Global DQN model;
 N_{agg} - Aggregation frequency
Output : Trained global model

```

1 foreach Communication round do
2   Get V2V payload size for each V2V link;
3   for  $t \leftarrow 1$  to  $T$  do
4     forall Agents do
5       Gets state observation  $s_t$  ;
6       Performs TrainingDQN( $t, s_t, T$ );
7     end
8   end
9   if  $N_{agg}$  is reached then
10    All agents upload local models to BS ;
11    /* Aggregate models as (13) */
12     $\theta^g \leftarrow \frac{1}{N_{agent}} \cdot \sum_{id=0}^{N_{agent}} \theta_{agent_{id}}$  ;
13    BS distributes  $\theta^g$  back to agents;
14  end

```

With the important aspects introduced, the proposed federated DRL-based algorithm for resource allocation is summarized in Algorithm 2, termed FL-MX for the essential building block of the FL and mixed DRL agent cooperation level.

V. EVALUATION CONFIGURATIONS

In this section, the experimental setup, benchmarks, and performance metrics are presented to evaluate the effectiveness of the proposed federated DRL algorithm in a single-cell cellular vehicular network. Simulations have been implemented using Python 3.11 with Numpy [36] and Pytorch [37]. Result visualization is generated using Matplotlib [38]. Note that all benchmarks and the proposed solution are trained and evaluated online using the wireless communication measurements from the implemented simulations.

A. Simulation Setups

The simulation setup is based on the evaluation methodology of the urban case as defined in [39]. This technical report includes detailed vehicle types and mobility models, V2V and V2N channel models, V2V data traffic models, etc. The main simulation parameters are summarized in Table I. Specifically, the simulation arena is set to contain four urban grids ($m \times m$) as defined in Table A-1, Annex A in [39] to focus on a local area centered at a cross-road intersection. Moreover, the network traffic model is based on Model 2 (medium traffic intensity) in Section 6.1.5 of [39]. Note that the configuration of V2N communications has been set according to the street canyon scenario [40]. As in [15], the discrete V2V power control in this work is set to have four options, i.e., [23, 10, 5, -100] dBm, where -100dBm represents “zero” V2V transmission power.

TABLE I
SIMULATION CONFIGURATIONS

Setup properties	Parameters and values
Simulation time	30 Seconds
Carrier frequency	6 GHz
Total bandwidth	140 MHz
Number of V2N links	5
Number of V2V links	5
Antenna height	BS: 10 m VUE: 1.5 m
Antenna gain	BS: 8 dBi VUE: 3 dBi
VUE drop & mobility model	Option A for urban case in [40]
Pathloss model	V2N: 3GPP Urban Micro V2V: 3GPP Urban case
Transmission power	BS: 24 dBm UE: [23, 10, 5, -100] dBm
Fast fading	Rayleigh fading
Shadowing	V2V: As in Table A.1.4-1 of [5] V2N: As in Table 7.4.1-1 of [41]
Receiver noise figure	BS: 5 dB VUE: 9 dB
Noise Power	-114 dBm
Network traffic model	Periodic traffic, 10 ms per packet arrival
V2V payload size	1200 bytes (20% probability), 800 bytes (80% probability)
V2V latency requirement	10 ms

TABLE II
NEURAL NETWORK HYPERPARAMETER CONFIGURATIONS

Hyperparameter	Value
Hidden layer number	3
Neuron size	$512 \times 512 \times 256$
Activation function	ReLU
Output Layer	Linear
Optimizer	AdamW
Learning rate	10^{-4}
Learning rate decay	0.97
τ	0.05

The neural network models are configured to be the same for all algorithms considered for a more controlled performance evaluation, including the DRL-based benchmarks introduced in Section V-B. These configurations are summarized in Table II. To be specific, rectified linear units (ReLU), i.e., $f(x) = \max(0, x)$, are used as the activation function for DQN. Moreover, the “Adam with decoupled weight decay” (AdamW) optimizer [41] is used to train the ANN parameters with an initial learning rate of 0.0001, and an exponential learning rate decay of 0.97 applied every episode. The replay buffer is configured to have the capacity of $2e^6$ with a mini-batch sample size of 1280 for the DQN for all considered benchmarks. For DRL benchmarks utilizing FL, the global aggregation frequency is set to every 30 seconds.

B. Benchmarks

The performance of the proposed FL-MX is compared with three established DRL-based benchmark methods in the literature, a fully distributed proposal by disabling the aggregation

of FL-MX models, and the baseline performance of a random strategy. These benchmarks are introduced as follows:

- *C-MARL*: This is the multi-agent DRL-based resource allocation scheme developed in [15]. C-MARL is implemented in a “centralized training, distributed execution” manner, so each agent takes its actions based on state observations using only local information, while a reward based on the V2V and V2N transmission rates of all agents is transmitted to the central controller for a joint reward calculation during training.
- *FL-CO*: This is the multi-agent DRL-based joint resource allocation and mode selection scheme developed in [23]. The allocation of resources in this scheme is extracted and applied to the problem formulated in (9) by modifying the action space.
- *Distributed*: This is the fully distributed multi-agent DRL algorithm derived from the proposed FL-MX by disabling the FL aggregation architecture so that all vehicular agents only learn on their local observations without any knowledge-sharing, representing the extreme RL situation of “independent learning”. All other setups, such as the input features and the reward design, are kept the same as FL-MX to achieve an ablation test.
- *Centralized*: This is a single-agent centralized solution developed in [14], where the network controller is modeled as the agent and the resource allocation decisions are made sequentially for each VUE. The inclusion of this benchmark gives a clear performance comparison between single-agent and multi-agent approaches in resource allocation.
- *Random*: This is the baseline that chooses the spectrum sub-band and V2V transmission power for each V2V link randomly, i.e., without any resource allocation optimization strategy.

C. Performance Metrics

For performance evaluation, all learning algorithms are first trained episodically in the same environment (initiated positions and trajectories of the vehicle, fading behaviors). After training, the algorithms are evaluated in testing environments whose data have not been experienced by the algorithms. The evaluation considers the following network performance metrics:

- 1) *V2V payload delivery ratio E* : This is the primary metric, as safety applications are more essential and critical for V2X applications as defined in (8) and $0 \leq E \leq 1$.
- 2) *Sum V2N transmission rate Λ* : This is the sum transmission rate of all V2N links defined in (5).
- 3) *Aggregated gain $\mathbb{G}$* : As the optimization problem is a max-max problem with two optimizations expressed by the above two metrics, an aggregated gain is defined to achieve a more grounded comparison, defined as:

$$\mathbb{G} = \frac{E_{drl} - E_{rnd}}{E_{rnd}} + \frac{\Lambda_{drl} - \Lambda_{rnd}}{\Lambda_{rnd}} \quad (14)$$

where E_{drl} and E_{rnd} stand for the V2V payload delivery ratio obtained by a considered DRL algorithm and the Random benchmark, respectively. The same applies to

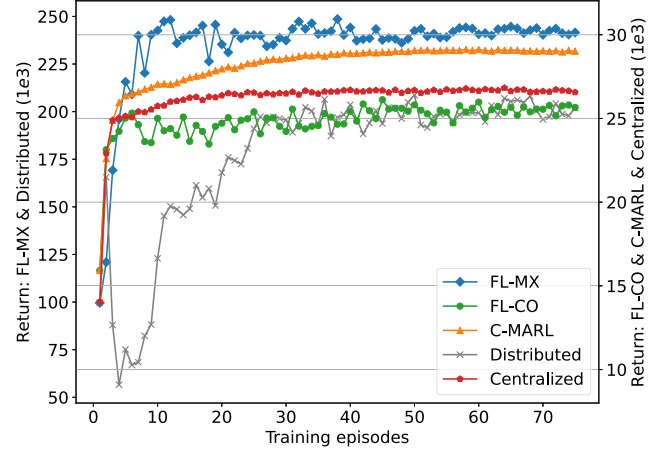


Fig. 3. Convergence Analysis: Cumulative return of considered DRL algorithms. FL-MX and Distributed share the same reward design, while C-MARL and FL-CO have another shared reward design comparable to that of the Centralized benchmark for controlled comparison. FL-MX converges quickly with high return, FL-CO reaches relative convergence the fastest yet at a local optimum, C-MARL has the most stable yet slowest convergence, and Distributed diverges before returning to a converging state.

Λ_{drl} and Λ_{rnd} for the sum V2N transmission rates. This metric indicates how much gain, considering both the above metrics, an algorithm can achieve compared with the Random benchmark’s baseline performance. Without losing the generalization ability, E and Λ are set to have the same weight in the summation of (14).

- 4) *Communication cost $\mathbb{C}$ and frequency $\mathbb{F}$* : Since the benchmarks require different information to be transmitted among agents or between agents and the FL aggregator, it is important to understand the communication overheads and communication frequency introduced for each scheme’s proper operation.

In addition, the convergence behaviors are also analyzed for the learning algorithms.

VI. EXPERIMENTAL RESULTS

To produce performance results, an offline training stage under the above configuration was first performed for all learning-based algorithms. After training, the algorithms are deployed in the same configuration for multiple runs with different VUE trajectories and mobility for direct performance evaluation.

A. Convergence Analysis

For machine learning algorithms, the convergence behavior will give valuable insight into how each algorithm learns the target task through the training process. For all DRL-based benchmarks considered and the proposed solution, the convergence trends are shown in Fig. 3. As stated above, the reward design has been set to be the same, where applicable, to guarantee a more controlled performance comparison of different agent cooperation levels.

Among all benchmarks, C-MARL and Centralized have the smoothest convergence trend, which has almost no fluctuation. The reason behind this for C-MARL is its cooperative and direct information-sharing design with all agents aiming to achieve

a common goal, although the convergence is relatively slow, and full convergence can only be claimed towards the end of training. For Centralized, only one agent learns to train the policy, making the overall learning process faster and more stable. In contrast, the Distributed benchmark initially has a diverging behavior until Episode 10, after which it becomes more convergent. This benchmark's behavior highly reflects the characteristics of unstable independent learning as discussed in the literature, although it manages to reconverge in this specific case.

For the proposed FL-MX and the benchmarking FL-CO, the overall convergence trends are similar, suggesting that the knowledge-sharing enabled by model aggregation helps FL-MX converge effectively compared to the Distributed benchmark with otherwise the same design. However, it is worth noting that FL-MX manages to converge after Episode 20, while FL-CO reaches a relative convergence at a much earlier Episode 10. Compared with C-MARL, the reason behind it may be that FL-CO exploits the benefit of knowledge sharing between both the shared-reward calculation and the FL model aggregation. Nevertheless, FL-CO reaches a lower plateau compared to C-MARL despite having the same reward function design, suggesting that the addition of FL model aggregation may have influenced the discovery towards the global optimum in exchange for fast convergence.

Overall, the proposed FL-MX has a balanced learning behavior with respect to both convergence speed and stability and the ability to reach the global optimum, suggesting its potential for further investigation in both theoretical proof and real-world implementation. A lesson is also learned that a more complex design, such as that of FL-CO, may require more dedicated algorithmic fine-tuning to take advantage of both cooperating agents and FL model aggregation.

B. Performance Evaluation Results

To obtain results from performance evaluation, all learning algorithms have been first trained using the same scenario following the configurations in Table I. After training, all considered algorithms are deployed with different configurations of the VUE trajectories, including those utilized for training, while other settings, such as the simulation parameters, are kept the same. For this performance evaluation, 10 rounds of simulations (with different trajectory configurations of VUEs) were performed to reduce the effect of variances for the evaluation.

1) *Performance Results With Varying Payload Sizes:* Figs. 4 and 5 show the average performances of V2V and V2N to increasing V2V payload sizes (N times the original payload size configured in Table I) for all considered algorithms in all 10 rounds of simulations. An obvious decreasing trend can first be found in both figures for all algorithms, as larger V2V payloads require more time to be fully transmitted with the same amount of resources, which reduces the overall successful V2V delivery ratio and V2N transmission rates.

Regarding the V2V delivery ratio E , Fig. 4 shows that FL-CO, C-MARL, and Centralized perform similarly. The proposed FL-MX stands out the best in maximizing E , outperforming all the

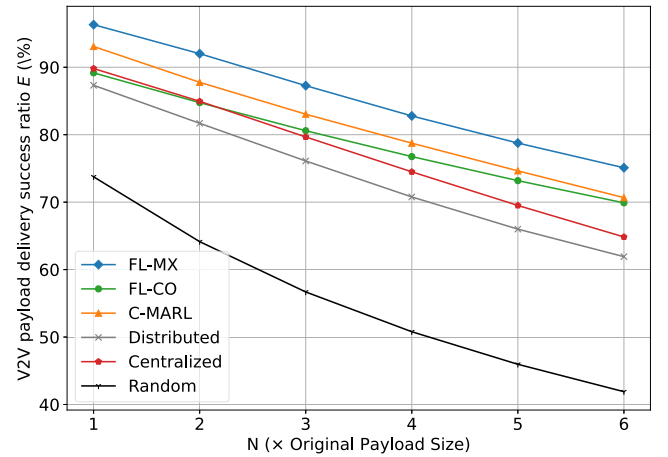


Fig. 4. System-wise V2V payload delivery ratio E under varying V2V payload sizes.

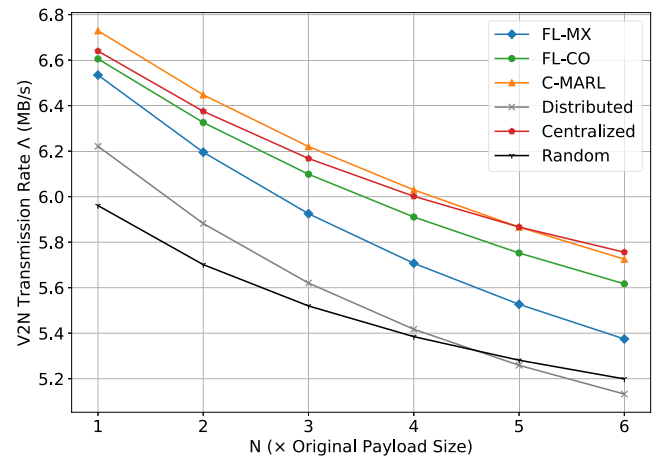


Fig. 5. System-wise V2N transmission rates Λ under varying V2V payload sizes.

above three benchmarks while having significant improvements compared to the Distributed benchmark ($\sim 11\%$ improvement) and the Random baseline ($\sim 40\%$ improvement) for all cases.

Regarding system-wise V2N transmission rates Λ , all learning algorithms except the Distributed benchmark see significant improvement compared to the random baseline. The proposed FL-MX's results are $\sim 7\%$ worse than those of the best-performing C-MARL on average. This difference becomes $\sim 6\%$ and $\sim 5\%$ compared to the inferior FL-CO and Centralized benchmarks. An interesting discovery is that the Distributed benchmark only has similar and even inferior Λ compared to the Random baseline, suggesting its suboptimal learning outcome by sacrificing too much V2N performance in exchange for E .

Combining both metrics, Fig. 6 shows the results of $\mathbb{G}$, as defined in (14). The Centralized and FL-CO benchmarks have worse $\mathbb{G}$ results as they failed to reach a higher E , although they improved to Λ . The proposed FL-MX has overall matching performance compared to the C-MARL benchmark.

2) *In-Depth Performance Under Default Payload Size:* For the same 10 rounds of simulations, this section dives deep into

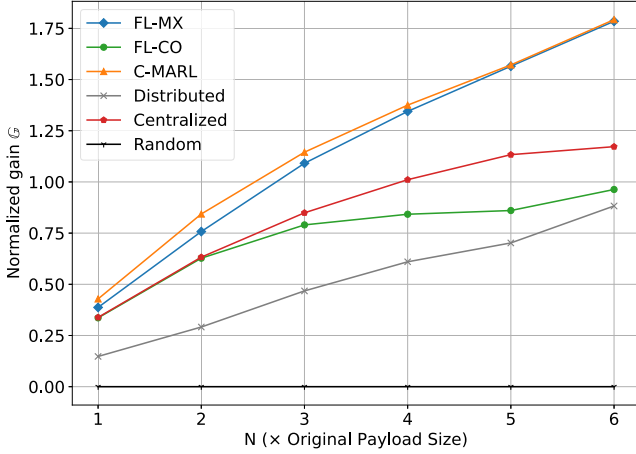
Fig. 6. Aggregated gain $\bar{G}$ results under varying V2V payload sizes.

TABLE III
SYSTEM-WISE E AND Λ RESULTS FOR ALL BENCHMARKS AND THE RANDOM
BASELINE FOR ALL TEST CASES

Algorithm	FL-MX	FL-CO	C-MARL	Distri.	Central.	Random
E (%)	96.31	89.17	93.08	87.33	89.81	73.76
Λ (MB/s)	6.53	6.61	6.73	6.22	6.64	5.96

the results for the default V2V payload size. Table III shows the average results of E and Λ for all benchmarks, which are the same as the results for $N = 1$ in Figs. 4 and 5.

Compared to the Random baseline, the proposed FL-MX achieves the highest E with a positive Λ gain, while C-MARL manages to have the highest Λ with satisfactory E . As expected from the convergence analysis for higher return throughout training, FL-MX and C-MARL have higher E and Λ compared to Distributed and FL-CO, respectively. Although FL-CO manages to converge the fastest, it only achieves intermediate performance results for both E and Λ due to reaching a local optimum during training, according to the convergence analysis above. Furthermore, since the Distributed benchmark recovered from diverging to relative convergence during training, it has the lowest E gain among all learning algorithms while also having similar Λ compared to the Random baseline, suggesting an obvious suboptimal learning outcome.

Fig. 7 shows the aggregated gain $\bar{G}$ results with the mean values, medians, and standard deviations identified for all the learning algorithms in the 10 test trials. Unlike the case of $N = 1$ shown in Fig. 6, which only shows the average $\bar{G}$ results, the proposed FL-MX achieves the highest median $\bar{G}$ with a 0.01 edge over C-MARL; FL-CO obtains the intermediate $\bar{G}$, while Distributed still gets a worse result compared to multi-agent approaches. The Centralized benchmark also fails considering both the average and median results, with an outlier that is very low $\bar{G}$. This shows the Centralized benchmark's overall under-exploration of the environment while learning, which sometimes generalizes poorly to other scenarios of the same environment. The same conclusion can be drawn from the Distributed benchmark.

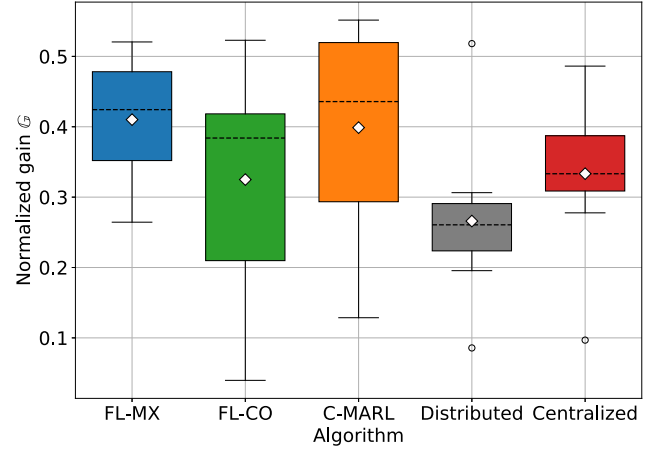


Fig. 7. The aggregated gain $\bar{G}$ results among all test rounds for different benchmarks. Dashed lines indicate the medians, while white diamond markers show the mean values for all algorithms. The proposed FL-MX has overall matched performance with C-MARL regarding the mean and median $\bar{G}$, but is more stable across different test trials, where C-MARL has a more polarized performance; the FL-CO and the Distributed benchmarks both have highly polarizing performances across the trials with lower mean and median $\bar{G}$.

Referring to the variances, FL-MX has the most consistent performance over all 10 trials, suggesting the algorithm's robustness in similar but changing scenarios. This outperforms C-MARL, whose simulation results are much more variant than FL-MX despite having a nearly matched median and mean $\bar{G}$. FL-CO, in comparison, has large variances, suggesting that it is less stable in generalization. The Centralized and Distributed benchmarks both have outliers, demonstrating much inferior robustness.

In summary, these results indicate that the proposed FL-MX under a mixed agent cooperation setting can match or even exceed the performance of the well-established multi-agent DRL algorithms with full agent cooperation, making such a design pattern promising for further explorations.

C. Generalization Tests With Newly Joined Agents

In this set of tests, an agent is configured not to have a trained model and thus resembles a newly joined agent to compare the generalization capability between the proposed FL-MX, FL-CO, and C-MARL. For FL-MX and FL-CO, the newly joined agent will receive a copy of the FL global model at the start of the simulation, while a copy of a trained model from another randomly chosen agent will be used for C-MARL. A new simulation is iterated through all 5 agents, imitating the newly joined agent with a vehicular trajectory configuration unseen in the previous experiments.

The results of $\bar{G}$ for the 5 agents are shown in Fig. 8 while the results of the V2V delivery ratio E and the V2N transmission rate Λ are summarized in Table IV. FL-MX stands out as the best-performing solution for the newly joined agent for the highest achieved $\bar{G}$ and E while also making improvements to Λ . The same holds for FL-CO, which comes best at Λ , while having second-best E . However, C-MARL has the least achieved gain in all three metrics. The reason behind this can be that another

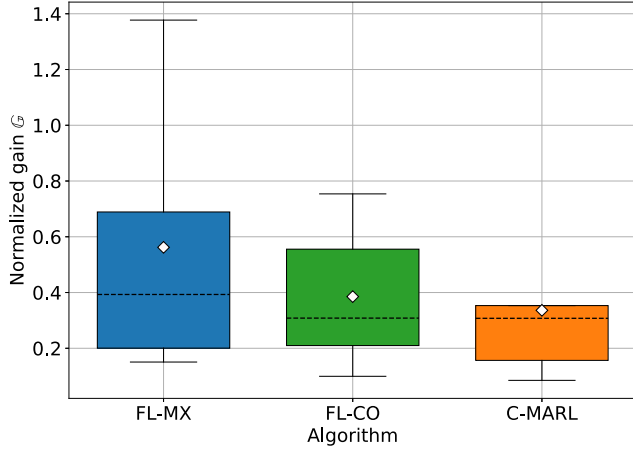


Fig. 8. Aggregated gain $\bar{G}$ results of the newly joined agents for the three considered benchmarks.

TABLE IV

AVERAGE E AND Λ RESULTS OF THE NEWLY JOINED AGENTS FOR THE THREE CONSIDERED BENCHMARKS AND THE RANDOM BASELINE

Algorithm	FL-MX	FL-CO	C-MARL	Random
E (%)	95.34	91.70	76.16	55.20
Λ (MB/s)	1.31	1.38	1.29	1.28

agent's local model does not have sufficient knowledge of the whole environment and thus only makes suboptimal decisions when the local model is adopted by the newly joined agent. This finding shows the effectiveness of the global model introduced by the FL architecture in benefiting potential new users who enter the considered environment.

D. Communication Costs and Frequency Analysis

The radio environment in vehicular networks is highly dynamic and vehicles frequently travel in and out of a local area. As a result, it is important for resource allocation schemes to adapt to the changing environment, which corresponds to online updating of ANN models for DRL algorithms. Such an online update requires information sharing, which leads to communication costs for the multi-agent DRL algorithms to further train the ANNs.

For the proposed FL-MX, new agents require downloading a copy of the global model from the aggregator at the beginning of the online update. For a given period T and aggregation frequency f_{agg} , the total communication cost of FL-MX can be derived as follows:

$$\mathcal{C}_{mx} = (T \cdot f_{agg} \times 2N_{vue} + N_{new}) \times S_{ann} \quad (15)$$

where N_{vue} and N_{new} represent the number of agents who participated in the online update and the number of new agents who require a copy of the global model, respectively, while S_{ann} represents the model size of the ANN model. Note that the communication frequency requirement $\mathbb{F}$ is the global aggregation frequency f_{agg} .

For C-MARL, the throughput information of V2V and V2N links from all agents must be sent to the central controller for a shared reward calculation and feedback to enable updates to

TABLE V
QUANTITATIVE DEMONSTRATION OF $\mathcal{C}$ AND $\mathbb{F}$, CALCULATED VIA (15)–(17) USING PARAMETERS FROM SIMULATION CONFIGURATIONS

Algorithm	FL-MX	FL-CO	C-MARL
$\mathcal{C}$ (kB/s)	255.2	290.12	34.92
$\mathbb{F}$ (Hz)	1/30 (f_{agg})	100 (f_{trans})	100 (f_{trans})

the online model. Assuming that the throughput information of both link types and the calculated reward have the same data size and are indicated as D_{tp} , the overall communication cost of C-MARL can be formulated as:

$$\mathcal{C}_{ma} = T \times (N_{vue} \times 3D_{tp} \times f_{trans}) + N_{new} \times S_{ann} \quad (16)$$

where it is assumed that N_{new} new agents managed to get a trained copy of ANN from a neighbor, and f_{trans} represents the transmission frequency of the throughput and joint reward information.

The FL-CO benchmark requires both federated model aggregation and local information sharing, resulting in a communication cost over T as:

$$\mathcal{C}_{co} = T \times (N_{vue} \times 3D_{tp} \times f_{trans}) + \mathcal{C}_{mx} \quad (17)$$

Note that for both C-MARL and FL-CO, the required communication frequency $\mathbb{F}$ is f_{trans} for the normal operation of both algorithms during online model updates.

As for the Distributed benchmark, since it is the extreme case of independent learning, no information is shared among the agents at all, and thus it has no communication cost or required communication frequency.

1) *Quantitative Case Study:* To give a numeric demonstration of the above analysis, on the implemented platform, S_{ann} is 696.0kB in storage and D_{tp} is 8 bytes for a 64-bit floating point number. Following the simulation configurations in Table I, f_{agg} and f_{trans} are set to 1/30Hz and 100Hz, respectively. T and N_{vue} are configured to be 30 seconds and 5 agents. Assuming one of the agents is newly joined and does not have a model, hence $N_{new} = 1$. The communication cost results are summarized in Table V for one round of simulation.

Note that S_{ann} is the main contributor to the overall large communication cost for FL-MX. However, FL-MX only requires such information to be transmitted every 30 seconds, as in the configuration, while C-MARL requires data transmission at each scheduling step, which is 10 ms, as in the configuration, and hence a much stricter requirement for real-time data sharing.

E. Discussions

In summarizing different aspects, the overall comparison among the benchmarks is shown in Fig. 9.

The proposed FL-MX exhibits matching performance and stable learning convergence compared to the state-of-the-art, fully cooperative multi-agent DRL solution, C-MARL. Using the federated learning architecture, FL-MX enables the design of a DRL algorithm with a mixed agent cooperation level that does not require frequent local information transmission, while facilitating simple model sharing for new agents. However, the ANN model size leads to relatively large communication costs, which can be a major aspect for improvement.

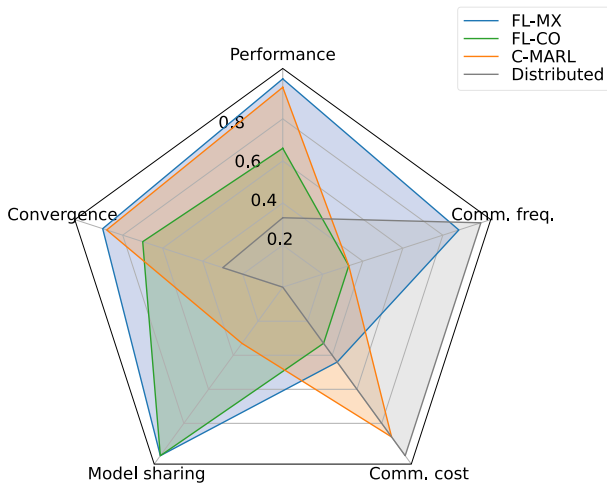


Fig. 9. The overall comparison among all algorithms. For all categories, the higher the value, the more advantageous for an algorithm. E.g., the proposed FL-MX has the overall best performance, convenient model sharing, and requires a lower communication frequency for model transmission, yet it has a higher communication cost due to the ANN model transmission requirement.

In comparison, although FL-CO has the highest level of information sharing via federated model aggregation and local information sharing, it is observed that it only presents the advantage of fast convergence during training. As the convergence results show that FL-CO reaches a local optimum, it does not achieve a satisfactory performance compared to those of C-MARL and FL-MX. Moreover, FL-CO has the highest communication cost and strictest communication frequency requirement due to the combination of both information sharing mechanisms, although it also has the advantage of simple model sharing, like FL-MX.

The Distributed benchmark does not require any information sharing due to agents' independent learning, and thus does not introduce any communication cost and frequency requirement. Although it manages to recover from a divergent trend during training, the Distributed benchmark does not have satisfactory performance on any considered measures, which is aligned with the reports in the existing literature.

Note that the communication cost of FL-MX is comparatively high and can lead to scalability issues for large-scale deployment, especially when considering online model updates. This remains true for other FL-based solutions. Research has started to tackle this issue of FL's deployment in vehicular networks, and promising techniques are being investigated to reduce FL's communication overheads. An example is to implement selective model transmission for model aggregation and to distribute the global model only to newly joined vehicle agents [42]. Other approaches involve dynamically fine-tuning FL's transmission code length to adapt to the requirements of model updates [43] or compressing deep learning models to reduce model size while maintaining performance [44]. With such techniques integrated, FL can evolve into a more practical solution on various deployment scales.

One more important aspect is that the proposed FL-MX yields a new information structure for multi-agent reinforcement learning with the federated learning architecture and FedAvg.

The general convergence of such a structure still requires theoretical support. Although the convergence of FL using collected datasets has been extensively studied [45], [46], extending these analyses to the context of multi-agent DRL still requires more attention from the research community, especially when multi-agent reinforcement learning itself is still grappling with theoretical convergence proofs [12].

VII. CONCLUSION

The dynamic radio environment and the differentiated QoS requirements in vehicular networks have led to the topic of optimized resource allocation for C-V2X, especially for an underlay sidelink configuration. For decentralized resource allocation in the Mode 2 setup, existing research focuses on a cooperative multi-agent reinforcement learning setting that achieves satisfying performance. However, such a design requires frequent transmission of local data, while the existing literature has hardly investigated other algorithmic designs with different levels of agent cooperation.

In this work, a federated DRL solution (FL-MX) is proposed for joint spectrum and power allocation in C-V2X systems with Mode 2 underlay sidelink setup. The proposed design is in a mixed agent-cooperation setting that does not require local observations to be shared among agents. Additionally, this work implements state-of-the-art solutions within fully cooperative learning settings, along with an extreme case of independent learning. The goal is to examine how different levels of agent cooperation affect the performance of the resource allocation scheme and its communication requirements. To evaluate performances, a latest standard-compliant simulation testbed is also developed for joint power & sub-band allocation experiments, where all benchmarks are set to have the same configurations when applicable for more controlled comparisons.

Through extensive simulation, the proposed FL-MX outperforms the best state-of-the-art solution with more stable generalization, achieving at least an improvement of 22.55% and 9.5% in the V2V payload delivery ratio and the V2N throughput compared to the Random baseline, respectively. Furthermore, FL-MX requires much less frequent data transmission, i.e., 30 seconds, compared with the 10 ms required by the state-of-the-art benchmark under the simulation configuration. However, due to the relatively large size of the ANN model, FL-MX requires more data transmission to update the online model, which requires further investigation for potential improvements. Overall, the evaluation results suggest a promising direction for future research to explore DRL-based solutions under mixed agent cooperation levels to avoid frequent direct information sharing, thus improving the practical aspects of real-world development.

For future work, real-world federated DRL deployment is an important research direction considering how communication-related factors influence the model aggregation process. For example, how unreliable communication channels influence the federated model aggregation and its effect on overall performance [47]. In addition, improving the DRL algorithm's generalization capability to various observation and action space sizes is also vital for potential real-world deployment. The potential

of adopting graph neural networks for this is being investigated, as the model has been found effective in the authors' previous work on a related task [48].

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